Working FIFO w/ 8 writers, each writing 64000, and 1 reader

```
(base) tunabeluga@AndrewGaming:~/ECE357/ps6/p2$ ./main -w 8 -n 64000
Beginning test with 8 writers, 64000 items each
Writer 1 completed
Writer 0 completed
Reader stream 1 completed
Writer 2 completed
Reader stream 0 completed
Reader stream 2 completed
Writer 3 completed
Reader stream 3 completed
Writer 4 completed
Reader stream 4 completed
Writer 5 completed
Reader stream 5 completed
Writer 6 completed
Writer 7 completed
Reader stream 6 completed
Reader stream 7 completed
All streams done
Waiting for writer children to die
All children exited normally
```

To break the synchronization, I removed the spin-lock mutex right before the critical section in fifo_wr, allowing multiple writers to update the tail. This creates a write–write race condition and corrupts FIFO ordering.

```c
void fifo_wr(struct myfifo *f, unsigned long d){
    sem_wait(&f->empty); //wait for at least 1 empty slot

    //COMMENTED OUT: while(sem_try(&f->mutex) == 0); //spin until we get the mutex

    //we're in the endgame now (critical region)
    f->buffer[f->tail] = d;
    f->tail = (f->tail + 1) % MYFIFO_BUFSIZ; // round it back

    // release the spin lock
    //COMMENTED OUT: sem_inc(&f->mutex);

    sem_inc(&f->full);
}
```

The result of the broken FIFO:

```
(base) tunabeluga@AndrewGaming:~/ECE357/ps6/p2$ ./main -w 8 -n 64000
Beginning test with 8 writers, 64000 items each
Detected out-of-sequence word 0:99 (expecting 98)
^C
```